Tema Geometrie

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$$B = \{f_1(1,1,1), f_2 = (1,1,-1), f_3 = (1,-1,-1)\}$$

$$e_1 = \frac{f_1}{\|f_1\|} = \frac{f_1}{\sqrt{\langle f_1, f_1\rangle}} = \frac{(1,1,1)}{\sqrt{1^2+1^2+1^2}} = \frac{1}{\sqrt{3}}(1,1,1)$$

$$e_2 = \frac{e_2'}{\|e_2'\|} = \frac{3}{2\sqrt{6}} \cdot \frac{1}{3}(2,2,-4) = \frac{1}{\sqrt{6}}(1,1,-2)$$

$$e_2' = f_2 - \langle f_2, e_1\rangle e_1 = (1,1,-1) - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}(1,1,1) = \frac{1}{9}(2,2,-4)$$

$$\|e_2'\|^2 = \frac{4+4+16}{9} = \frac{24}{9} = \frac{8}{3} \Rightarrow \|e_2'\| = \frac{2\sqrt{2}}{\sqrt{3}} = \frac{2\sqrt{6}}{3}$$

$$e_3' = f_3 - \langle f_3, e_1\rangle e_1 - \langle f_3, e_2\rangle e_2 = (1,-1,-1) +$$

$$\frac{1}{\ } \cdot \frac{1}{\ }(1,1,1) - \frac{2}{\ } \cdot \frac{1}{\ }(\ )$$

$\sqrt{3}\ \sqrt{3}\ \ldots\ \sqrt{6}\ \sqrt{6}\,(1,1,2) = \frac{1}{3}(4,-2,-2) - \frac{1}{3}(1,1,-2) =$

$\frac{1}{3}(3,-3,0) = (1,-1,0)$

$\|e_3'\|^2 = 2 \Rightarrow \|e_3'\| = \sqrt{2}$

$\tilde{B} = \left\{ e_1 = \frac{1}{\sqrt{3}}(1,1,1),\ e_2 = \frac{1}{\sqrt{6}}(1,1,-2),\ e_3 = (1,-1,0)\frac{1}{\sqrt{2}} \right\}$

$f_3 = \alpha(1,-1,0),\ \alpha \in \mathbb{R}^*$ CP: $\alpha = 1 \Rightarrow (1,-1,0)$

$B = \{f_1, f_2, f_3\} \xrightarrow{\text{POG-S}} \tilde{B}$

$e_3 = \frac{f_3}{\|f_3\|} = \frac{1}{\sqrt{2}}(1,-1,0) = e_1 = \frac{f_1}{\|f_1\|} = \frac{1}{3}(2,2,1)$

$e_2 = \frac{e_2'}{\|e_2'\|} = \frac{5}{9} \cdot \frac{9}{30}(-1,-1,4) = \frac{1}{10}(-1,-1,4)$

$e_2' = f_2 - \langle f_2, e_1 \rangle e_1 = (-1,-1,2) + \frac{2}{3} \cdot \frac{1}{3}(2,2,1) =$

$$\frac{1}{9}(-5,-5,20)$$

$$\|e_2'\|^2 = \frac{1}{81}(25+25+400) = \frac{450}{81} = \frac{50}{9}$$

$$W = (-1,1,2) = W_1 e_1 + W_2 e_2 + W_3 e_3$$

$$[W]_{\tilde{B}} = (W_1, W_2, W_3)$$

$$\langle W, e_1\rangle = W_1\langle e_1, e_1\rangle + W_2\langle e_2, e_1\rangle + W_3\langle e_3, e_1\rangle = W_1$$

$$W_1 = \langle W, e_1\rangle = \frac{1}{\sqrt{3}}(-1+1+2) = \frac{2}{\sqrt{3}}$$

$$W_2 = \langle W, e_2\rangle = \frac{1}{\sqrt{6}}(-1+1-4) = \frac{-4}{\sqrt{6}}$$

$$W_3 = \langle W, e_3\rangle = \frac{1}{\sqrt{2}}(-1-1+0) = \frac{-2}{\sqrt{2}}$$

$$\Rightarrow [W]_{\tilde{B}} = \left(\frac{2}{\sqrt{3}}, -\frac{4}{6}, \frac{-2}{\sqrt{2}}\right)$$

$$Q(x) = x_1^2 + x_2^2 + x_3^2 + \dots$$

$Q = x_1 + 3x_2 + x_3^2 - 2x_1x_2 - 4x_2x_3$

met Gauss + Jacobi

G: $Q(x) = x_1^2 + 3x_2^2 + x_3^2 - 2x_1x_2 - 4x_2x_3$

$= (x_1 - x_2)^2 - x_2^2 + 3x_2^2 + x_3^2 - 4x_2x_3$

$= (x_1 - x_2)^2 + 2(x_2 - x_3)^2 - x_3$

$y_1 = x_1 - x_2 \qquad y_2 = x_2 - x_3 \qquad y_3 = x_3$

$Q(x) = y_1^2 + 2y^2 - y_3^2 \rightarrow$ f. canonică

$z_1 = y_1 \qquad z_2 = \sqrt{2}\, y_2 \qquad z_3 = y_3$

$Q(x) = z_1^2 + z_2^2 - z_3^2 \rightarrow$ f. canonică normală

$r = \operatorname{rg} Q = 3$

$p = 2$

$q = 1$

$s = p - q = 1$

J: $G_{B_0} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \end{pmatrix}$

$$\begin{pmatrix} 0 & -2 & 1 \end{pmatrix}$$

$\Delta_1 = 1 \neq 0$

$$\Delta_2 = \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 3 - 1 = 2 \neq 0$$

$$\Delta_3 = \begin{vmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{vmatrix} = 3 - 4 - 1 = -2 \neq 0$$

$\Delta_1, \Delta_2, \Delta_3 \neq 0 \Rightarrow$ Th. Jacobi

$\exists$ o bază B' în $\mathbb{R}^3$ a.î. $Q(x) = \frac{1}{\Delta_1}(x_1')^2 + \frac{\Delta_1}{\Delta_2}(x_2')^2 + \frac{\Delta_2}{\Delta_3}(x_3')^2$

$Q(x) = \frac{1}{1} x_1'^2 + \frac{1}{2} x_2'^2 - x_3^2 = x_1'^2 + \frac{1}{2} x_2'^2 - x_3'^2$ forma canonică

$y_1 = x_1'$ $\quad y_2 = \frac{1}{\sqrt{2}} x_2'$ $\quad y_3 = x_3'$

$Q(x) = y_1^2 + y_2^2 - y_3^2 \Rightarrow$ f canonică normală

$r = rg\, Q = 3$

$p = 2$

$q = 1$

$s = p - q = 2 - 1 = 1$

$F(x, y) = 2x_1y_1 + x_2y_2 - 2x_1x_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2$

$F(x, x) = 2x_1^2 + x_2^2 - 2x_1x_2 - 2x_2x_1 - 2x_2x_3 - 2x_3x_2 = Q(x)$

Gauss + Jacobi

G: $Q(x) = 2x_1^2 + x_2^2 - 4x_1x_2 - 4x_2x_3$

$= (x_1 - x_2)^2 + x_1^2 - 2x_1x_2 - 4x_2x_3$

$= (x_1 - x_2)^2 + (x_1 - x_2)^2 + x^2 - 4x_2x_3$

$= 2(x_1 - x_2)^2 + (x_2 - 2x_3)^2 - 4x_3^2$

$y_1 = x_1 - x_2 \qquad y_2 = x_2 - 2x_3 \qquad y_3 = x_3$

$2x_3 \qquad y_3 = 2x_3$

$Q(x) = 2y_1^2 + y_2^2 - y_3^2 \rightarrow$ f. canonická

$z_1 = \sqrt{2}\, y_1 \qquad z_2 = y_2 \qquad z_3 = y_3$

$Q(x) = z_1^2 + z_2^2 - z_3^2 \rightarrow$ f. canonická normalá

$r = \operatorname{rg} Q = 3$

$p = 2$

$q = 1$

$s = p - q = 1$

7: $G_{B_0} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$

$\Delta_1 = 2 \neq 0$

$\Delta_2 = \begin{vmatrix} 2 & -2 \\ -2 & 1 \end{vmatrix} = 2 - 4 = -2 \neq 0$

$$\Delta_3 = \begin{vmatrix} -2 & 1 & \\ 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{vmatrix} = -8 \neq 0$$

$\Delta_1, \Delta_2, \Delta_3 \neq 0 \Rightarrow$ Th Jacobi

$\exists$ o bază B' în $\mathbb{R}^3$ a.î. $Q(x) = \frac{1}{\Delta_1}(x_1')^2 + \frac{\Delta_1}{\Delta_2}(x_2')^2 + \frac{\Delta_2}{\Delta_3}(x_3')^2$

$Q(x) = \frac{1}{2} x_1'^2 - x_2'^2 + \frac{1}{4} x_3'^2 \rightarrow$ f. canonică

$y_1 = \frac{1}{\sqrt{2}} x_1'$ $\quad y_2 = x_2'$ $\quad y_3 = \frac{1}{2} x_3'$

$Q(x) = y_1^2 - y_2^2 + y_3^2 \rightarrow$ f canonică normală

$r = \operatorname{rg} Q = 3$

$p = 2$

$q = 1$

$$D = p - q = 1$$